A3. Epidemic Spreading on Complex Networks: The SIS Model

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Complex Networks

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1 Introduction

The study of epidemic spreading in complex networks allows us to understand how the structure of interactions between individuals influences the persistence and reach of a disease. Unlike the SIR model, where individuals gain immunity, the Susceptible-Infected-Susceptible (SIS) model describes infections like the common cold or certain computer viruses, where individuals can be reinfected immediately after recovery.

The primary objective of this report is to analyze the stationary state of an SIS dynamics—defined by the fraction of infected nodes ρ—across different network topologies. We focus on comparing two fundamental types of networks:

- **Erdös-Rényi (ER):** A random network where connections are distributed homogeneously.
- **Barábasi-Albert (BA):** A scale-free network characterized by a power-law degree distribution and the presence of high-degree nodes or "hubs."

By varying the infection probability λ and the recovery probability μ, we aim to identify the epidemic threshold and validate our stochastic Monte Carlo simulations against the theoretical predictions of the Microscopic Markov Chain Approach (MMCA).

2 Implementation and Design Decisions

To ensure the simulation was both accurate and consistent with the course requirements, the following decisions were made:

- **Software Choice:** Python was used for all simulations. Although it is generally slower than compiled languages, we chose to maintain consistency with the course labs and utilize the standard *NetworkX* and *NumPy* libraries for network analysis.
- **Performance and Reliability:** To optimize execution time, we implemented vectorized operations using NumPy. Additionally, the number of repetitions was set to 50. This combination allowed us to mitigate Python's slower processing speed while ensuring enough statistical data to produce reliable results for a 1000-node system.
- **Initialization and Stochastic Fade-out:** Every simulation started with 50 infected nodes. This decision was crucial to prevent "stochastic fade-out," where an outbreak might accidentally disappear in the first few steps if the initial infected nodes recover before they can spread the disease to their neighbors.

3 Methodology

3.1 The SIS Epidemic Model

We implemented the Susceptible-Infected-Susceptible (SIS) model, where individuals do not acquire permanent immunity. At each discrete time step, nodes transition between two states:

- **Recovery:** An infected node returns to the susceptible state with a recovery probability μ.
- **Infection:** A susceptible node becomes infected by its neighbors. If a node has k_{inf} infected neighbors, its probability of being infected is given by:

$$P_{inf} = 1 - (1 - \lambda)^{k_{inf}} \tag{1}$$

  where λ is the infection probability per contact.

3.2 Network Topologies

The dynamics were tested on four synthetic networks with $N = 1000$ nodes:

- **Erdős-Rényi (ER):** Constructed with a connection probability $p = \langle k \rangle/(N - 1)$ to achieve average degrees of $\langle k \rangle = 4$ and $\langle k \rangle = 6$.
- **Barabási-Albert (BA):** Generated using preferential attachment with parameters $m = 2$ and $m = 3$ to achieve average degrees of approximately $\langle k \rangle = 4$ and $\langle k \rangle = 6$, respectively.

3.3 Simulation Procedure

The Monte Carlo simulations were performed using a synchronous update rule. At each time step t, the state of all nodes is recorded, and a copy is used to determine the transitions for $t + 1$. This ensures that nodes updated early in the loop do not influence their neighbors until the following time step.

To characterize the stationary state, we calculated the average infection density ρ:

$$\rho = \frac{1}{100} \sum_{t=901}^{1000} \tag{2}$$

$$\rho = \frac{1}{T} \sum_{t=T_{trans}}^{T_{max}} \frac{I(t)}{N} \tag{3}$$

This value is further averaged over 50 independent realizations to minimize stochastic fluctuations and ensure the reliability of the steady-state results.

3.4 Microscopic Markov Chain Approach (MMCA)

To provide a theoretical baseline, we utilized the MMCA. This approach describes the probability p_i of a node i being infected through a fixed-point iteration:

$$p_i = (1 - q_i)(1 - p_i) + (1 - \mu)p_i \tag{4}$$

The term q_i represents the probability of node i escaping infection from all of its neighbors simultaneously. The theoretical equations are then solved by repeatedly updating these probabilities for every node until the values stabilize.

We consider the system to have reached a solution when the average change between iterations is extremely small (below a tolerance of 10^{-6}). Once these individual probabilities are fixed, the final theoretical prevalence is calculated as the average across all nodes:

$$\rho = \frac{1}{N}\sum_{i=1}^{N} p_i \tag{5}$$

This value represents the steady-state fraction of infected individuals predicted by the MMCA theory, which we then compare against our simulation data.

4 Simulation Setup

To ensure the reproducibility of our results, all simulations were performed using a consistent set of parameters designed to capture the steady-state behavior of the SIS model. The experimental configuration is summarized below:

- **Population Size:** All networks were constructed with $N = 1000$ nodes.
- **Infection Probability:** λ was varied within the range $[0.0, 0.3]$ using a step size of 0.01.
- **Recovery Probabilities:** Simulations were conducted for two specific recovery rates: $\mu = 0.2$ and $\mu = 0.4$.
- **Initial Conditions:** To avoid stochastic extinction in the early stages, each simulation was initialized with 50 randomly infected nodes ($\rho(0) = 0.05$).
- **Temporal Parameters:** Each realization ran for a total of $T_{max} = 1000$ time steps. A transient period of $T_{trans} = 900$ steps was discarded to ensure the system reached the stationary state.
- **Statistical Averaging:** For every combination of λ and μ, we performed 50 independent realizations ($N_{rep} = 50$). The final prevalence $\langle\rho\rangle$ represents the average over the last 100 time steps across all 50 repetitions.

5 Results

5.1 Monte Carlo Simulations

The results of the simulations, presented in Figures 1 and 2 , illustrate how the stationary infection fraction ρ behaves as a function of the infection probability λ. In all cases, we observe a phase transition: for small values of λ, the system remains in a disease-free state ($\rho = 0$), but once a critical epidemic threshold is crossed, the infection persists in the population.

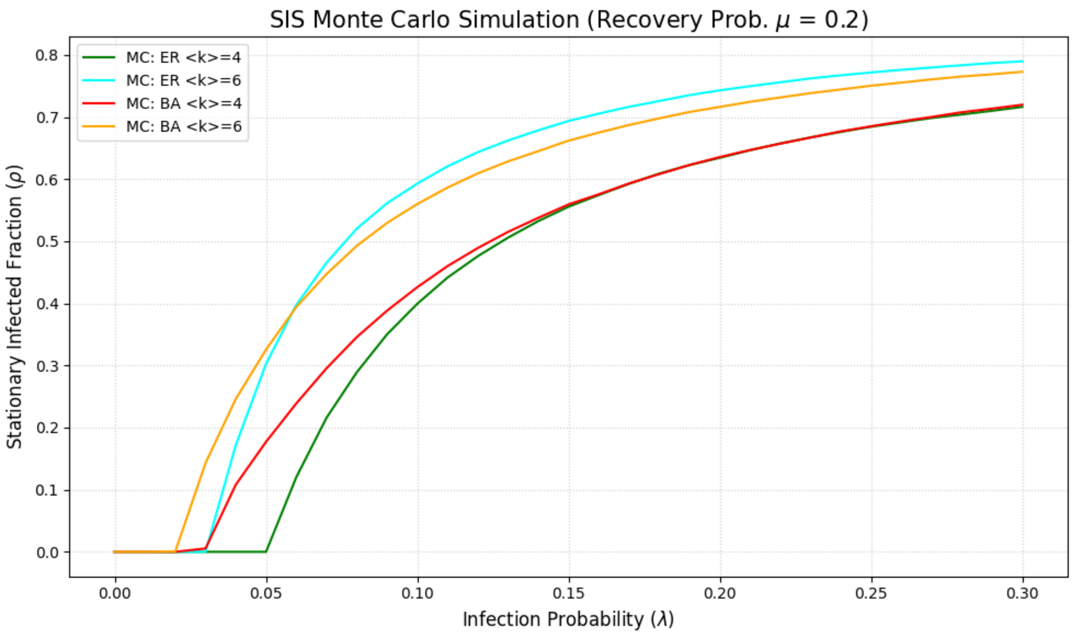

Figure 1: Stationary infected fraction ρ vs. infection probability λ for a recovery rate $\mu = 0.2$.

Several key observations can be made from these results:

- **Impact of Topology (ER vs. BA):** As seen in Figure 1, the Barabási-Albert (BA) networks reach the epidemic threshold earlier than the Erdős-Rényi (ER) networks for the same average degree. This is due to the presence of high-degree **hubs** in the BA structure, which act as "super-spreaders" that sustain the infection even when the individual infection probability is low.
- **Influence of Network Density ($\langle k \rangle$):** Increasing the average degree from $\langle k \rangle = 4$ to $\langle k \rangle = 6$ significantly lowers the epidemic threshold and increases the final prevalence ρ. This confirms that a more connected population allows the disease to spread more efficiently.

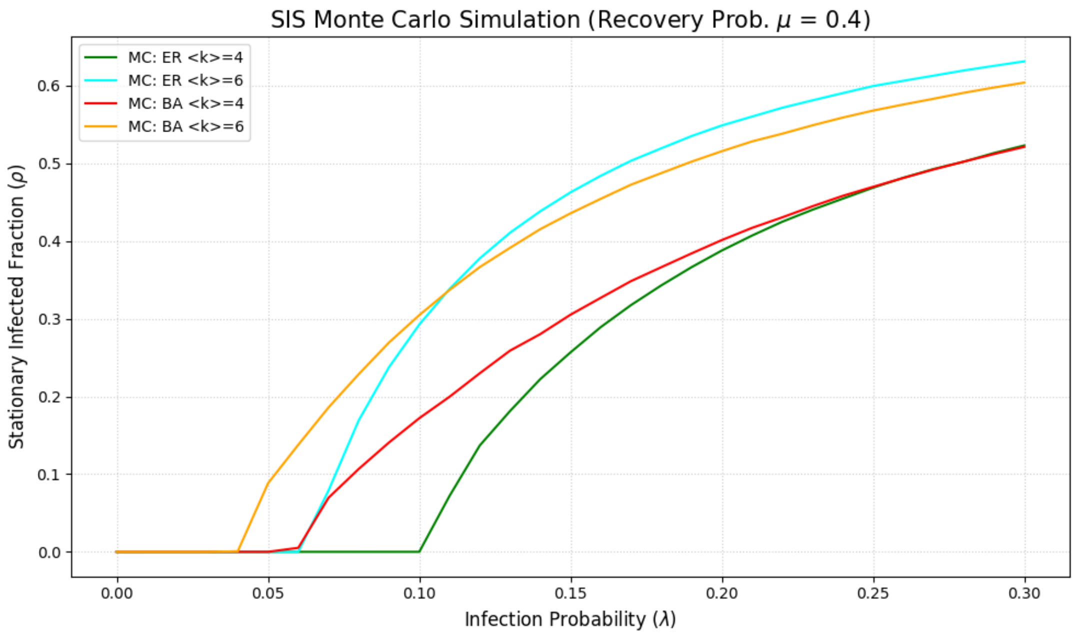

Figure 2: Stationary infected fraction ρ vs. infection probability λ for a recovery rate $\mu = 0.4$.

- **Effect of Recovery Rate (μ):** Comparing Figure 1 ($\mu = 0.2$) and Figure 2 ($\mu = 0.4$), we see that a higher recovery rate shifts the entire curve to the right. When individuals recover faster, a much higher infection probability λ is required for the disease to survive in the stationary state. For example, in the ER network with $\langle k \rangle = 4$, the threshold shifts from approximately $\lambda \approx 0.05$ to $\lambda \approx 0.10$ when μ is doubled.

These results highlight that while the recovery rate μ and the number of contacts $\langle k \rangle$ determine the scale of the outbreak, the underlying network architecture is the fundamental driver of how early an epidemic begins.

5.2 Theoretical Comparison: MC vs. MMCA

To validate the accuracy of our simulations, we compared the Monte Carlo (MC) results with the theoretical predictions obtained through the Microscopic Markov Chain Approach (MMCA). As shown in Figures 3 and 4, there is a very high degree of agreement between the two methods across all network topologies and recovery rates.

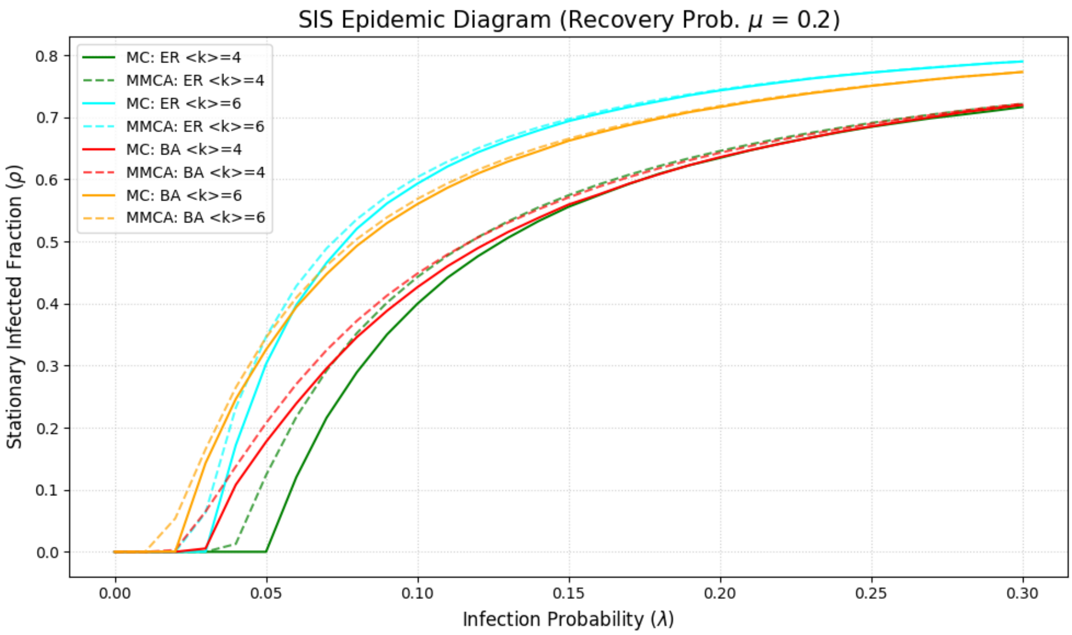

Figure 3: Comparison between MC simulations (solid lines) and MMCA theory (dashed lines) for $\mu = 0.2$. The theory accurately captures the prevalence for all four network configurations.

The comparison reveals several important theoretical insights:

- **General Accuracy:** The MMCA successfully predicts the stationary infected fraction ρ in the endemic phase (the curved part of the plots). This confirms that the fixed-point iteration correctly approximates the long-term behavior of the SIS model.
- **Threshold Discrepancy:** A small but visible discrepancy occurs near the epidemic threshold. In both figures, the MMCA (dashed lines) tends to predict the onset of the epidemic at a slightly lower λ than what is observed in the Monte Carlo simulations.
- **Independence Assumption:** This overestimation of the epidemic's strength is a known characteristic of the MMCA. It occurs because the theory assumes that the states of neighboring nodes are independent, whereas in a real simulation, correlations exist (e.g., if node A is infected, it is more likely that its neighbor B is also infected).

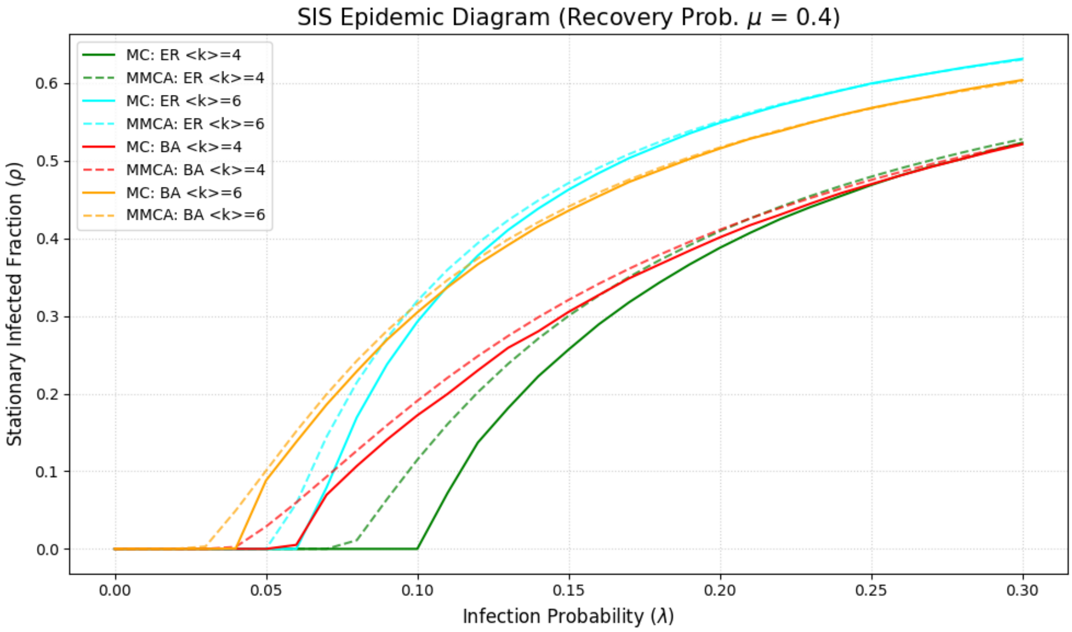

Figure 4: Comparison for $\mu = 0.4$. Even with a higher recovery rate, the MMCA remains a robust predictor of the steady-state prevalence.

In conclusion, while the MMCA slightly overestimates the vulnerability of the network near the critical point, it serves as an excellent and computationally efficient tool for predicting the overall dynamics of disease spreading in complex structures.

6 Discussion

6.1 ER vs. BA Network Dynamics

The primary difference between the spreading dynamics observed in this experiment lies in the structural topology of the networks. Erdős-Rényi (ER) networks are characterized by a homogeneous degree distribution, where most nodes possess a number of connections close to the average $\langle k \rangle$. In contrast, Barabási-Albert (BA) networks are scale-free and heterogeneous, featuring a small number of highly connected "hubs" and a large number of poorly connected peripheral nodes.

This structural difference fundamentally alters the epidemic threshold λ_c. In our simulations, the BA networks consistently transition to the endemic state at a lower infection probability than the ER networks. This behavior is explained by the relationship between the threshold and the moments of the degree distribution:

$$\lambda_c \approx \frac{\mu \langle k \rangle}{\langle k^2 \rangle} \tag{6}$$

In BA networks, the presence of hubs significantly increases the second moment $\langle k^2 \rangle$, which mathematically drives the epidemic threshold toward zero. Physically, these hubs act as catalysts that capture and redistribute the infection across the network, sustaining the disease even when the individual probability of transmission λ is very low.

Furthermore, the growth and saturation patterns of the prevalence ρ vary between the two models. While the BA networks are more vulnerable to an initial outbreak due to their hubs, the ER networks often reach a higher total prevalence at very high values of λ. This occurs because the uniform connectivity of the ER model ensures that once the infection probability is high enough to clear the threshold, the disease can easily reach almost every node in the system. In BA networks, the peripheral “leaf” nodes are only connected via specific paths through the hubs; if the recovery rate μ is sufficiently high, these nodes can remain healthy even if the core of the network stays infected, leading to a slightly lower saturation point compared to the more compact ER structure.

7 Conclusions

The implementation and analysis of the SIS model across different topologies demonstrate that network structure is a primary driver of epidemic spread. Our results confirm that the presence of high-degree hubs in Barabási-Albert networks significantly lowers the epidemic threshold compared to Erdős-Rényi networks, making heterogeneous populations more vulnerable to outbreaks even when the individual infection probability is low.

Furthermore, we observed that increasing the average degree $\langle k \rangle$ accelerates the transition to the endemic state, while a higher recovery rate μ effectively suppresses the disease by shifting the threshold toward higher infection probabilities. These findings underscore the critical role of both local node dynamics and global network architecture in determining the long-term prevalence of a disease.

Finally, the comparison between methods shows that the Microscopic Markov Chain Approach (MMCA) provides a robust theoretical upper bound for the stationary state observed in stochastic simulations. While the MMCA tends to slightly overestimate the infection density near the critical threshold due to its assumption of node independence, it serves as an exceptionally accurate and computationally efficient tool for predicting epidemic behavior in large-scale complex networks.